Commitment-Aware Liquidity-Shortfall Prediction Using Traditional Machine Learning

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ABSTRACT: This study evaluates whether explicit recurring-commitment features improve prediction of next-calendar-month negative net cash flow using traditional machine learning. The experiment uses a synthetic transaction benchmark containing 11,365 records from 249 synthetic users and a derived panel of 3,444 monthly snapshots from 241 eligible users. Two HistGradientBoostingClassifier models were compared: a 27-feature class-blind baseline built from income, aggregate outflows, net-cash-flow history, transaction reliability and simple forecast proxies; and a 41-feature model adding 14 commitment-specific features derived from pre-labelled subscription, EMI/loan and BNPL transaction classes. Both models used the same chronological training, validation and test periods. On the 422-observation test set, Average Precision increased from 0.9564 to 0.9617 (+0.0053), while ROC-AUC increased from 0.9446 to 0.9561 (+0.0115). Balanced accuracy increased from 89.20% to 90.07% (+0.87 percentage points), and Brier score improved from 0.0834 to 0.0745 (–0.0089). However, the 95% user-clustered bootstrap confidence intervals for all metric differences spanned zero (AP difference 95% CI: [–0.0038, +0.0151]; ROC-AUC difference 95% CI: [–0.0051, +0.0272]; balanced accuracy difference 95% CI: [–1.22, +2.98] percentage points). The results therefore do not establish a statistically significant or robust predictive improvement from the added commitment features on this benchmark. The target is negative monthly net cash flow, not observed overdraft, legal default or insolvency. Because the data are synthetic and transaction classes are pre-labelled, further validation on complete, de-identified real-world statements is required.

Keywords: liquidity-shortfall prediction, recurring financial commitments, financial transactions, gradient boosting, temporal validation, synthetic data

NOMENCLATURE

Symbol	Description	Unit / range
I_t	Successful income observed in month t	Currency units
C_t	Successful outflows assigned to commitment classes	Currency units
V_t	Other successful non-income outflows	Currency units
CF_t	Monthly net cash flow: I_t – C_t – V_t	Currency units
Y_{t+1}	Binary indicator for negative net cash flow in the next calendar month	0 or 1
p_i	Predicted probability of Y=1 for test observation i	[0, 1]

I. INTRODUCTION

Personal financial management systems commonly summarize historical transactions and provide spending categories or budget forecasts. Short-term financial planning, however, also depends on the relationship between future inflows and outflows that have already become recurring obligations. Examples include digital subscriptions, equated monthly instalments (EMIs), and buy-now-pay-later (BNPL) repayments. Predicting a future negative cash-flow month from transaction records is therefore different from classifying individual transactions: it requires a time-aware view of income and expenditure, while also determining whether explicitly represented recurring commitments add information beyond aggregate spending history.

Prior studies demonstrate the feasibility of transaction categorization, recurring-pattern extraction, cash-flow forecasting and transaction-based credit-distress prediction [1]–[7]. These studies address related but different prediction targets and datasets. The present study focuses on an incremental-feature question: when a conventional model already uses aggregate income, expenses, net cash flow and payment reliability, does adding explicit commitment-specific information improve prediction of a negative net-cash-flow outcome in the following calendar month?

Research question

Does incorporating explicitly modelled recurring financial commitment features improve the predictive performance of a traditional machine-learning model for next-calendar-month negative net cash flow compared with a baseline built from class-blind aggregate financial features?

Contributions

- A controlled comparison of two HistGradientBoostingClassifier models using the same chronological partitions, with 27 class-blind baseline features and 14 additional commitment-specific features.
- An evaluation of ranking, threshold-based classification and probabilistic error, with paired user-clustered bootstrap confidence intervals rather than unsupported p-values.
- An implementation audit that aligns the feature lookback window and baseline definition with the research question, and an explicit account of what the synthetic benchmark can and cannot establish.

II. RELATED WORK

A. Transaction categorization and personal-finance analytics

García-Méndez et al. developed a two-stage banking-description classifier combining a short-text similarity filter with a support vector machine, emphasizing that merchant descriptions are brief, noisy and domain-specific [4]. Lesner et al. described personalized transaction categorization at large scale for business accounting, using information shared across users to classify transactions even when counterparties are new [5]. These studies establish transaction categorization as a mature component, but neither evaluates whether commitment-specific features improve prediction of next-month negative net cash flow.

Kotios et al. combined rule-based transaction categorization and machine learning with a cash-flow forecasting component for small and medium-sized enterprises [3]. Yang et al. proposed a deep-learning system jointly addressing personal transaction classification and budget prediction [7]. The former concerns business accounts and the latter uses a deep-learning architecture; both differ from the present study in target formulation and experimental focus. Their relevance is that prediction can be coupled to upstream transaction understanding, while the additional value of explicit commitment features remains a separate empirical question.

B. Recurring financial patterns

Ibrain, Hernández and Peinado proposed unsupervised filtering approaches for discrete financial-event series, intended to remove noise and isolate recurring patterns before predicting the next transaction date or amount [2]. Lorpunmanee and Kamonsantiroj introduced CP-growth for mining partial-periodic recurring patterns across inter-transactions in time series [6]. The latter is a general pattern-mining method rather than a direct bank-payment classifier. These studies motivate temporal recurrence analysis, but neither is a direct baseline-versus-commitment-feature experiment for negative monthly net cash flow.

C. Transaction-based financial distress prediction

Monteux et al. studied transaction-only machine learning for short-term credit distress and overdraft-related outcomes under a structural shock using a large European bank dataset [1]. That work is directly relevant to the use of transactions as predictors of future financial distress, but its target is overdraft dynamics rather than the synthetic next-month net-cash-flow label used here. It also does not evaluate the incremental contribution of the 14 commitment-specific features examined in this study.

D. Positioning of the present study

Within the seven supplied papers, the authors did not identify a directly comparable experiment that isolates the incremental value of commitment-specific features for this particular next-calendar-month negative-net-cash-flow target. This is a bounded observation about the reviewed corpus, not a claim that no prior system has studied the problem. The paper's contribution is therefore an empirical ablation question, not a new machine-learning algorithm and not a claim of universal novelty.

III. PROBLEM FORMULATION AND DATASET

A. Target definition

Let I_t denote successful income transactions in calendar month t, C_t successful outflows assigned to the dataset classes Subscription, EMI/Loan or BNPL, and V_t all other successful non-income outflows. Monthly net cash flow is defined as:

$$CF_t = I_t – (C_t + V_t)$$

For a feature month t, the binary target is the observed net cash flow in the next calendar month t+1:

$Y_{t+1} = 1$ if $CF_{t+1} < 0$; otherwise $Y_{t+1} = 0$.

This operational target measures a negative monthly net cash flow. It is not an account-balance threshold, an observed overdraft, or proof that the user is unable to meet payments. The source dataset contains no true running balance or overdraft-limit field. The expression “liquidity shortfall” is therefore used as shorthand for the stated negative-net-cash-flow

proxy.

B. Dataset and panel construction

The supplied benchmark contains 11,365 synthetic transactions from 249 synthetic users dated 30 December 2023 through 31 December 2025. The transaction table has 18 columns, no nulls in the audit profile, and no exact duplicate rows. Class counts are Subscription 4,768, EMI 3,716, BNPL 1,281 and Normal 1,600. After monthly aggregation and requiring six prior calendar months plus a next-month target, the modelling panel contains 3,444 snapshots from 241 eligible users, with feature months from May 2024 through November 2025. All modelling findings are conditional on this simulated data-generation process.

Monthly rows without transactions are reindexed to zero in the pipeline. Consequently, a target month with no transaction rows becomes CF=0 and is labelled non-shortfall. This occurred in 224 of the 3,444 snapshots overall and in 30 of the 422 test snapshots. The analysis preserves the supplied simulation convention, but its interpretation depends on treating absent monthly transaction rows as complete zero activity rather than missing statement data.

C. Chronological data partitions

Partition (feature month)	Period	Snapshots	Users	Positive labels	Prevalence
Training	May 2024–May 2025	2,463	238	1,569	63.70%
Validation	June–August 2025	559	197	298	53.31%
Test	September–November 2025	422	161	200	47.39%
Full panel	May 2024–November 2025	3,444	241	2,067	60.02%

Table 1. Chronological modelling partitions. User counts overlap across time partitions; the test evaluates future periods for known users.

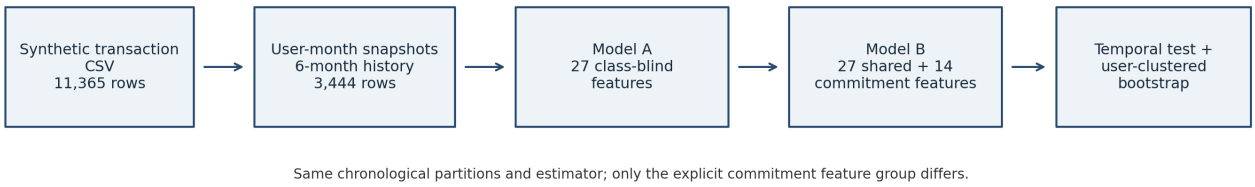


Fig. 1. Audited experimental workflow. Only the commitment-specific feature group differs between models.

IV. METHODS

A. Feature-set comparison

Model A is a 27-feature conventional baseline. Its features are computed without grouping transactions by commitment class or merchant category: six income-history features; seven aggregate-outflow and generic transaction features; seven aggregate net-cash-flow and buffer features; four recent transaction-status reliability features; and three baseline forecast proxies. The baseline includes historical total expenditure and net cash flow, so it does not discard the effect of recurring outflows from aggregate cash-flow history; rather, it does not explicitly disaggregate their amounts and types.

Model B retains all 27 Model A features and adds 14 commitment-specific features derived from the benchmark's pre-labelled transaction_class field using the recent six-month window. These include previous committed outflow, three- and six-month average commitments, commitment-to-income ratio, average subscription/EMI/BNPL burden, total recurring burden, expected committed expense, count of months with commitments, commitment growth, commitment-class diversity, and two compound forecast proxies. The implementation does not include explicit next-due-date or days-until-payment features, nor does it evaluate the accuracy of the upstream commitment detector.

Feature group	Model A	Model B	Description
Income history	6	6	Rolling means, slope, volatility and stability
Aggregate expenses / generic transaction statistics	7	7	Total outflows; no commitment-category grouping in shared baseline features
Net cash flow and buffers	7	7	Historical aggregate net cash flow
Transaction reliability	4	4	Recent success, failure, pending and resolved-payment rates
Baseline forecast proxies	3	3	Income, total expenses and net cash flow
Commitment levels and burden	0	9	Historical commitment amounts and class-specific burdens
Commitment dynamics and diversity	0	3	Active months, recent growth, class diversity
Commitment-aware compound forecasts	0	2	Total-expense and net-cash forecast proxies
Total	27	41	Model B adds 14 features

Table 2. Feature architecture after the audit of the original feature construction.

B. Feature calculations and leakage controls

The monthly transaction history was restricted to information available at the feature month t. Historical income and expenditure summaries use the preceding six calendar months; three-month aggregates use the latest three months. Commitment burdens are average monthly successful positive outflows within the same six-month lookback. In the audited run, baseline features were rebuilt from total outflows and generic transaction statistics so they do not depend on transaction_class-based commitment grouping. Model B uses the dataset's pre-labelled transaction_class values only for the added commitment features. The experiment therefore tests the value of commitment-aware feature enrichment conditional on those labels; it does not test whether the system can infer the labels reliably from raw bank descriptions.

The target is calculated from transactions in month t+1, while feature construction is restricted to months at or before t. The train/validation/test split is made by feature month, so outcome months do not overlap between partitions: training feature months end in May 2025, validation uses June–August 2025, and testing uses September–November 2025. This is a temporal forecast for known users, not a test on unseen users. Median imputation and scaling are fitted only on the training partition. The estimator is unweighted; the original script's sample-weight expression evaluated to one for both classes because the positive class was the training majority.

C. Classifier and hyperparameters

Both feature sets use scikit-learn HistGradientBoostingClassifier [8], [9] with learning_rate=0.06, max_iter=180, max_leaf_nodes=15, min_samples_leaf=25, l2_regularization=1.0 and random_state=42. The probability threshold for confusion-matrix metrics is 0.50. The model is evaluated with identical train/validation/test snapshots, and both model variants receive the same training protocol.

D. Evaluation metrics and uncertainty

Average Precision (AP) is the primary ranking metric because it emphasizes precision–recall performance at the observed class prevalence [10]. Secondary measures are ROC-AUC, accuracy, balanced accuracy, precision, recall, F1-score, Brier score and expected calibration error (ECE). The Brier score is the mean squared error between predicted probabilities and binary outcomes [11].

Paired 95% confidence intervals were computed from 1,000 user-clustered bootstrap resamples. At each iteration, test users are sampled with replacement and all their observed test-month rows are retained together, preserving within-user dependence. Confidence intervals are percentile intervals for the Model B minus Model A difference. The bootstrap fraction of resamples favouring a model is descriptive and is not treated as a conventional p-value. These intervals condition on the fitted models and saved test predictions; they do not quantify model-fitting or training-sample variability. No p-values are reported in this manuscript.

V. RESULTS

A. Test-set comparison

Table 3 reports the audited controlled experiment. Model B's AP is 0.0053 higher than Model A's (0.9617 vs. 0.9564), but its user-cluster bootstrap 95% confidence interval for the difference spans zero ([–0.0038, +0.0151]). ROC-AUC is higher by 0.0115 (0.9561 vs. 0.9446), also with an interval spanning zero ([–0.0051, +0.0272]). At the fixed 0.50 threshold, balanced accuracy is 90.07% for Model B versus 89.20% for Model A (+0.87 percentage points; 95% CI [–1.22, +2.98] percentage points), precision is 92.97% versus 90.58% (+2.39 percentage points), and recall is 86.00% versus 86.50% (–0.50 percentage

points). Model B's Brier score improves from 0.0834 to 0.0745 (–0.0089; 95% CI [–0.0190, +0.0012]), and empirical expected calibration error (ECE) is 0.0284 versus 0.0647 (–0.0363). While point estimates show modest directional gains, every 95% user-clustered confidence interval spans zero, confirming that incremental value cannot be claimed as statistically robust.

Metric	Model A	Model B	Δ (B–A)	95% user-cluster bootstrap CI for Δ
Average Precision (primary)	0.9564	0.9617	+0.0053	[–0.0038, +0.0151]
ROC-AUC	0.9446	0.9561	+0.0115	[–0.0051, +0.0272]
Accuracy	0.8934	0.9028	+0.0095	—
Balanced accuracy	0.8920	0.9007	+0.0087	[–0.0122, +0.0298]
Precision	0.9058	0.9297	+0.0239	—
Recall	0.8650	0.8600	–0.0050	—
F1-score	0.8849	0.8935	+0.0086	[–0.0141, +0.0307]
Brier score (lower is better)	0.0834	0.0745	–0.0089	[–0.0190, +0.0012]
ECE (lower is better)	0.0647	0.0284	–0.0363	—

Table 3. Test metrics from the audited feature construction (N = 422; 200 positive and 222 negative outcomes).

For Average Precision, ROC-AUC, balanced accuracy, F1 and Brier score, the reported intervals were independently recomputed from the unrounded saved test predictions using 1,000 paired user-clustered bootstrap resamples. In every case, the 95% confidence interval spans zero. Descriptively, the fraction of resamples with positive differences was 88.8% for AP, 93.4% for ROC-AUC, 78.5% for balanced accuracy, 76.6% for F1, and 95.8% for Brier improvement (4.2% positive delta). As documented in Section IV.D, these fractions are descriptive bootstrap proportions and are not p-values. The evidence does not establish a statistically significant improvement.

B. Threshold-based errors

Model	TN	FP	FN	TP
Model A: 27 features	204	18	27	173
Model B: 41 features	209	13	28	172

Table 4. Confusion matrices at the fixed probability threshold of 0.50.

At the default decision threshold of 0.50, Model B reduces false alarms by five cases (13 false positives versus 18 for Model A), yielding an improvement in precision from 90.58% to 92.97%. However, it incurs one additional false negative (28 versus 27), slightly lowering recall from 86.50% to 86.00% (172 versus 173 true positives out of 200 positive cases). Specificity improves from 91.89% to 94.14%. Thus, Model B operates with a slightly more conservative positive classification profile at the 0.50 threshold, reducing false alarms without a substantial penalty to detection.

C. Exploratory feature ablation

Variant	Features	AP	ROC-AUC	Balanced acc.	Brier
A: baseline	27	0.9564	0.9446	0.8920	0.0834
A + commitment projections	30	0.9598	0.9516	0.8937	0.0770
A + burdens and diversity	32	0.9545	0.9430	0.8882	0.0856
A + commitment history/ratios	33	0.9604	0.9572	0.9102	0.0716
B: full 41 features	41	0.9617	0.9561	0.9007	0.0745

Table 5. Predefined feature ablation on the same test partition. These variant comparisons are exploratory; confidence intervals were not calculated for every pairwise ablation.

Table 5 presents the exploratory ablation results on the test partition. Across subsets, the 33-feature variant adding commitment history and burden ratios achieves the highest balanced accuracy (91.02%) and lowest Brier score (0.0716), while the full 41-feature Model B achieves the highest point AP (0.9617). Permutation importance analysis on Model B identifies commitment_ratio (+0.0744 ± 0.0106) and previous_committed (+0.0230 ± 0.0051) as the two most influential commitment features, followed by baseline signals income_volatility (+0.0064) and previous_net_cash (+0.0042). Crucially, because these ablation subsets were evaluated post-hoc on the test partition without a separate held-out confirmation split, the 91.02% balanced accuracy must be interpreted strictly as an exploratory finding rather than confirmatory evidence of superiority.

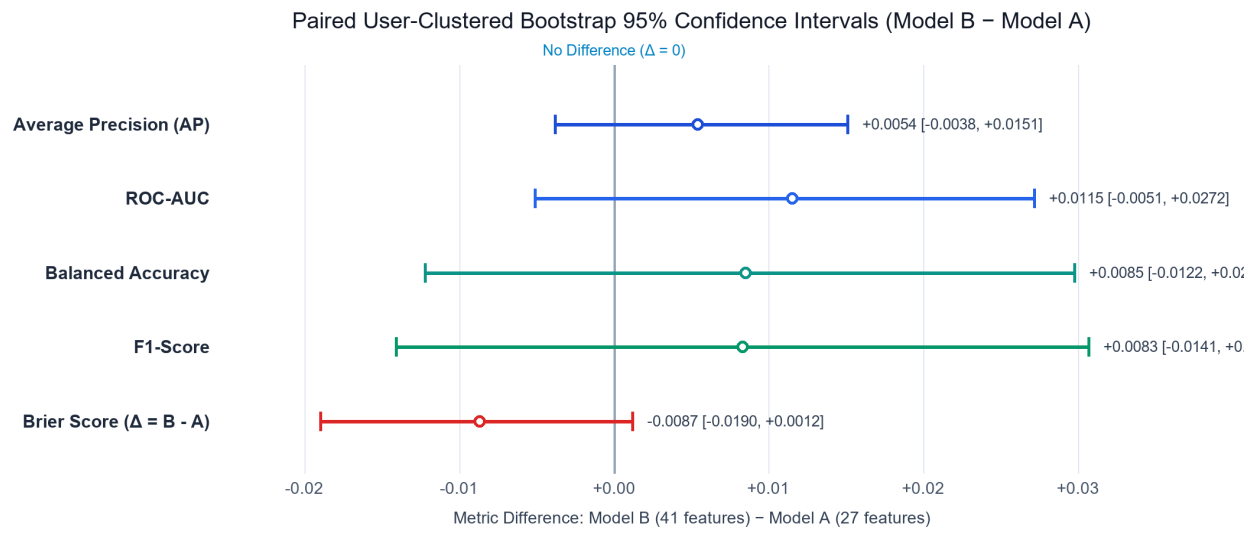


Fig. 2. Paired user-cluster bootstrap 95% confidence intervals for selected metric differences. Each interval crosses zero.

VI. DISCUSSION

The principal empirical finding is that explicit commitment features provide modest point improvements across ranking and discrimination—raising Average Precision by +0.0053, ROC-AUC by +0.0115, balanced accuracy by +0.87 percentage points, and reducing Brier score by –0.0089—yet every 95% user-clustered bootstrap confidence interval spans zero. Consequently, the hypothesis that commitment-aware modeling provides a statistically significant predictive advantage over conventional financial history is not supported at the standard 5% significance level. While Model B achieves higher precision by eliminating five false positives, it does so within a margin that remains consistent with sample variability.

The experiment also illustrates why the baseline definition matters. In the initially supplied runner, several transaction-level rates and class-specific burden aggregates used all available historical transactions while dividing some amounts by only a recent six-month denominator. The initial baseline also derived “variable” spending features and category-dispersion features using commitment-class information. For the research comparison, these were revised: transaction-level rates and commitment features now use the recent six-month window, and the 27 shared baseline features use aggregate outflows and generic transaction statistics without commitment-class or merchant-category grouping. Consequently, the results in this paper are from the audited rerun and intentionally supersede the preliminary result table in the original experiment package.

The updated baseline still includes historical aggregate net cash flow, which naturally reflects all types of outflows. Model B’s incremental input is not the existence of spending in the past; it is the explicit decomposition of recent outflows into labelled recurring commitment groups and related burden/dynamics features. That is the specific contribution evaluated here.

VII. LIMITATIONS AND THREATS TO VALIDITY

Synthetic benchmark and proxy target: all 11,365 transactions are simulated, and the label is next-month negative net cash flow—not an observed overdraft, account-balance breach, insolvency or inability to pay. Real-world performance is untested.

Pre-labelled commitments: Model B uses the dataset transaction_class labels for feature construction. The upstream detector is not evaluated, and the experiment includes no explicit next-payment-date or days-until-payment feature.

Missing months: user-months without transaction rows are reindexed to zero; 30 test target months have no transaction rows. This assumes complete coverage rather than missing statements and requires safeguards in real deployments.

Known-user temporal test: users overlap across train and test months. Results concern future periods for users with earlier history, not unseen-user generalization. The test contains 422 snapshots from 161 users over three feature months.

Threshold and uncertainty: results use a fixed 0.50 threshold. Cost-sensitive thresholding and post-hoc calibration were not studied. Bootstrap intervals condition on fitted models and do not include model-fitting variability; ECE has no interval.

Environment: the audited rerun was verified under Python 3.12, scikit-learn 1.9.0, pandas 3.0.5, NumPy 2.5.2, and SciPy 1.18.0. Reported metrics are tied to the included pipeline script and pinned environment.

VIII. CONCLUSION AND FUTURE WORK

On this synthetic transaction benchmark, the controlled comparison of a class-blind 27-feature baseline and a 41-feature commitment-aware model demonstrated modest positive point differences in Average Precision (+0.0053), ROC-AUC (+0.0115), balanced accuracy (+0.87 percentage points), and Brier score (–0.0089), alongside an empirical reduction in false alarms from 18 to 13. However, all paired user-clustered bootstrap 95% confidence intervals spanned zero. The experiment therefore demonstrates that explicit commitment features do not provide a statistically significant or robust incremental benefit over conventional aggregate cash-flow signals on this benchmark. Future work must validate these models on complete, de-identified real-world banking statements with verified labels, evaluate statement coverage safeguards, test generalization to unseen cold-start users, and explore cost-sensitive decision thresholds and payment-timing features.

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